STAT 24620=FINM 34700, STAT 32950
Multivariate Data Analysis
Lecture 8: CCA, multivariate regression, and RRR

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Outline

Lecture 7 recap: what CCA was trying to do

Given two centered random vectors $X \in \mathbb{R}^p$ and $Y \in \mathbb{R}^q$, CCA finds linear combinations

$$U = a^\top X, \qquad V = b^\top Y$$

that maximizes correlation.

$$\max_{a,b} \ \mathrm{Corr}(a^\top X, b^\top Y)$$

subject to variance-normalization and orthogonality constraints for later pairs.

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- CCA is **symmetric** in the two blocks.
- It answers: *which linear summaries of X and Y are most strongly associated?*
- It does *not* directly optimize prediction error of Y from X.

Preprocessing matters for CCA

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Practical tip:

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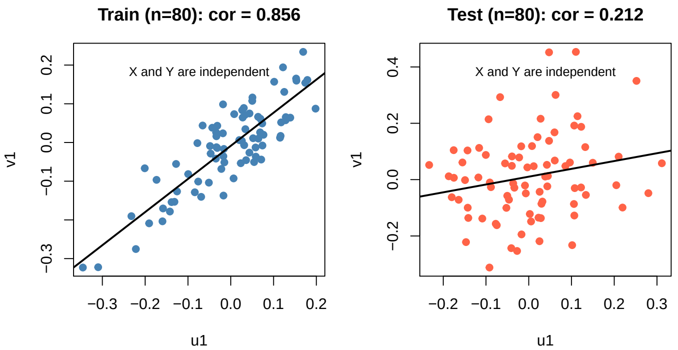
Practical tip:

- center the data and check for outliers;
- assess stability of canonical correlations and directions;
- be mindful of multicollinearity and small sample issues.

Overfitting in CCA: a simple example



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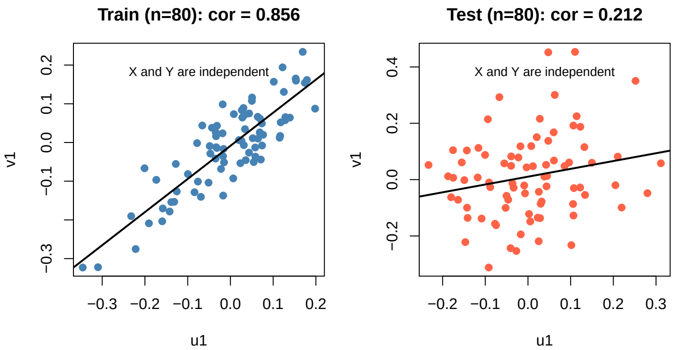

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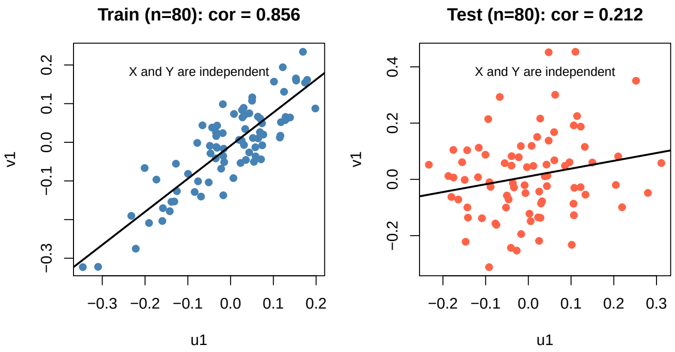

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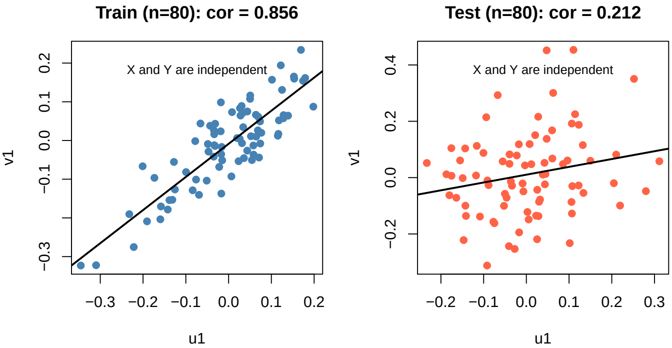

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CCA can find spurious relationships that do not generalize.

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Takeaway: large canonical correlation $\neq$ real relationship.

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Key point: statistical significance $\neq$ practical importance.

An R demo of CCA

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Question: What if our goal is to predict Y from X?

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- X plays the role of input/predictor;
- Y plays the role of output/response;
- predictive fit matters.

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Interpretation: each column of Y is predicted from the same predictor block X.

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Same ingredients, but different goals: association vs. prediction.

Connection: when Y is one-dimensional

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Conclusion:CCA and regression produce the same direction when Y is scalar.

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CCA finds low-dimensional structure in association; multivariate regression uses all directions for prediction.

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Reduced-Rank Regression (RRR): combine dimension reduction (like CCA) with prediction (like regression).

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Low-dimensional prediction $\Longleftrightarrow$ low-rank coefficient matrix.

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Solution:

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- Keep top r: $\hat{Y}_{\mathrm{RRR}} = U_r D_r V_r^\top$

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How to compute reduced-rank regression

$$\min_{B:\ \mathrm{rank}(B) \leq r} \|Y - XB\|_F^2$$

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RRR = best low-rank approximation of $\hat{Y}_{\mathrm{OLS}}$.

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RRR $=$ PCA applied to the fitted values $\hat{Y}_{\text{OLS}}$ from X to Y, not to raw Y nor X.

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RRR focuses on prediction; CCA focuses on scale-free association.

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- you want dimension reduction tailored to supervised prediction.

Prefer CCA when:

- the two blocks play symmetric scientific roles;
- the main goal is interpretation of shared structure;
- scale-normalized association is more important than prediction error.

RRR v.s. multivariate regression

Multivariate regression

$$
\begin{tikzcd}
 & Y_1 \\
X \arrow[ur] \arrow[r] \arrow[dr] & Y_2 \\
 & Y_3
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- separate models for each response

Reduced-rank regression

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RRR improves prediction by exploiting shared structure in Y.

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Goal: estimate out-of-sample prediction error for each rank r.

K**-fold cross-validation:**

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4. Average error across folds.

Final step:

- repeat for different r;
- choose r with smallest CV error.

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RRR is especially useful in high dimensions, but often needs regularization.

Live R notebook: comparing CCA with multivariate regression and RRR

Notebook file: `Lecture8_demo.html`, part II

Lecture 8 summary

1. CCA studies shared linear association between two variable blocks.
2. Multivariate regression predicts Y from X and is inherently asymmetric.
3. Reduced-rank regression imposes a low-dimensional predictive structure on the coefficient matrix.
4. The right method depends on whether the scientific question is about association, prediction, or both.